Tit4Tat Trading Bot: live demonstration guide

How to run the demo, what to say, the strategies we use and the tools we built with. Every command here was tested on 30 Oct 2026.

1. Live demo plan

Setup, 15 minutes before

- Laptop on the charger, sleep turned off, connected to the venue Wi-Fi so the platform (192.168.65.157) is reachable.
- Ask the organisers to reset your cash.
- Open three windows side by side: **(1)** a terminal for the bot, **(2)** the dashboard in your browser, **(3)** the platform website with the leaderboard.
- Start the dashboard early. It only reads the bot's log files, so it adds no load on the platform:

```
streamlit run dashboard.py
```

Demo, about 4 minutes

Step	What to do	What to point at
1	Start the bot: `python run_bot.py`	It loads the 40 stocks and their history, reads the stage, acknowledges heartbeats, then shows **ready=True**.
2	Let the days run	Each line: date · day N · 40 ticks · 400 ms · wealth. One full decision in under half a second.
3	Point at an order line	*FILLED BUY PRGD … weighted by volume flow (+0.18) over 10 stocks: 10.8% here*. Every trade states its reason: this is the explainable part.
4	Switch to the dashboard	Wealth curve, positions, recent orders with reasons, today's signals, news.
5	Switch to the website	The leaderboard wealth matches the bot's own number.
6	Mention the kill switch (don't trigger it)	A file named **STOP** means the bot only sells from then on; **FLATTEN** sells everything.

Backup if the platform isn't live during your slot

Replay of the 500 recorded days. It takes 26 seconds and ends at ₹14,94,245 with the leaderboard's columns:

```
python backtest.py --replay-live
```

The test suite. It takes 80 seconds and prints "OK" for 96 tests. Start it, then keep talking while it runs:

```
python -m unittest discover -s tests
```

`python verify.py` runs the full 21-point checklist, but it takes 2.5 minutes. Run it before your slot and show the result.

Don't do these during the demo

× Don't run a second bot, `paper_live.py` or `verify.py --live` while the platform is live. Everyone shares one IP, and too many requests gets the team blocked.

× Don't run plain `backtest.py` or `--episodes`. The history folder now mixes the old 10-stock and new 40-stock data, so the multi-year window gives misleading numbers. `--replay-live` is not affected.

× Don't press Ctrl+C in the bot terminal while it's trading.

2. What to convey: five messages

› **It's fully autonomous.** It reads Kafka, decides and trades with no human input.

› **It explains itself.** Every order is logged with a plain-English reason.

› **It's backed by evidence.** We tested 30+ ideas over 32 back-tested games and kept only what won.

› **It's built for real costs.** The scoring includes fees and Indian tax, so the bot plans its tax.

› **It's robust, and we're honest about risk.** The pre-flight results below show both.

Pre-flight measure	Value
Result	**+107.9%** (₹20.79 L) vs the market's +81.5%
Orders	148 sent, 140 filled, 2 rejected, 6 stopped by our own checks before sending
API requests	4,772, with 0 rate-limited and 0 errors
Headlines processed	38,904
Average decision time	458 ms
Deepest fall	**41%**, in the 2022 sell-off

3. Strategies we used

Adopted

- **Volume-weighted core:** 99% invested in the best 10 stocks by volume flow (buying minus selling volume over 50 days). Where buyers dominate, the stock gets more money.
- **Smart entry:** a stock is bought only while its volume flow is positive. Anything still waiting is bought by day 20.
- **Hold through dips with a concentration cap:** no stock may exceed 30%. Above that it's trimmed back to 26%, and 35% is the hard limit.

- **Risk guard:** an 8% drawdown halves new buying. There are also hard drawdown and daily-loss limits, plus the STOP / FLATTEN kill switch.
- **Tax planning:** use the ₹1 L tax-free long-term gain each year (sell long-term winners and buy them back at once, 15–31 March and in the last 5 days); tax-loss harvest (book losses to cancel taxable gains, then buy the same stock back); prefer selling shares held more than 365 days (10% tax instead of 15%).
- **Bear hedge:** short selling only on stocks outside the core, only when 30% or fewer stocks are in an uptrend, and closed within 5 days. It fired 10 times in the pre-flight test.
- **News:** a finance word-list scores each headline, and the score fades over time. It feeds the stock score but never triggers a trade on its own, because tests showed no next-day edge.

Tested and rejected

Trailing stops, RSI exits, rebalancing, selling on heavy-volume down days, weighting calmer stocks more, momentum rotation, mean reversion, buying in daily steps, and buying the dip. Each lowered the average after-tax return over the 32 back-tested games.

4. What we used

Area	Details
Language and libraries	Python 3 (Anaconda), kafka-python, requests, python-dotenv, pandas, Streamlit (dashboard)
Platform	Kafka topics market.ticks and market.news (consumer group group-mn2602); REST Order Gateway: orders, portfolio, tax report, stage, heartbeat
Indicators	EMA (fast, slow, trend), RSI 14, ATR 14, efficiency ratio, momentum, z-score, volume flow (50 days), news score
Safety in trading	Every order checked against the platform's rules before sending; request pacer (12 requests per 5 s); records checked against the server's
Staying connected	Kafka watchdog reconnects if the stream stalls; heartbeat worker acknowledges every heartbeat
Evidence and testing	Backtester that replays past runs and scores 32 back-tested games; verify.py 21-point checklist; 96 automated tests
Records and security	Every decision logged in JSONL files; token kept only in .env; submission ZIP scanned for secrets